How CDEF works

Methodology and algorithm justification for our replication of *Underwater Scene Enhancement via Adaptive Color Analysis and Multispace Fusion* (Yuan, Zhang, Cai, IEEE Journal of Oceanic Engineering, 2025)

In one paragraph

CDEF (Color Discrimination multispace Exposure Fusion) is a traditional, non-learning method for enhancing underwater photos. It first decides whether a photo is **greenish or bluish**, then **repairs the damage** water has done (too dark, wrong colours), then makes **four differently enhanced versions** of the repaired photo (two contrast stretches, one sharpened, one dehazed), and finally **blends the four** pixel by pixel, keeping from each version the parts that look best. The blend is done with image pyramids in the YUV colour space so that it has no visible seams or halos.

The problem it solves

Light travelling through water is changed in three ways that a normal camera cannot undo:

- **Absorption is colour dependent.** Red is absorbed within a few metres, then green, while blue travels furthest. Photos turn blue or green and lose their reds.
- **Scattering** by particles adds a veil of haze (back-scatter) and blurs edges (forward scatter). Contrast and detail drop, especially with distance.
- **Less light overall**, so photos are dark and noisy, and artificial lights create bright spots next to dark areas.

No single operation fixes all of these, and each fix has a side effect (a strong contrast stretch creates noise, dehazing over-darkens, colour correction can make the image grey). CDEF's central idea is to **compensate first, then produce several partial fixes and let a fusion step choose, per pixel, which fix to trust**. This follows the fusion approach of Ancuti et al., which CDEF extends with a colour-type decision, luminance compensation, two more inputs and a YUV fusion.

The three phases

Phase	Steps	Output
1. Degradation compensation (DC)	Tone test (Eq. 3); luminance compensation (BIMEF); colour compensation in CIELab (Eqs. 1-2)	One repaired image C
2. Fusion preparation (FP)	Two tone maps F1 (Eqs. 4, 6) and F2 (Eqs. 5, 6); on the tone-matched one, CBM sharpening gives F3 and IDE dehazing gives F4	Four inputs F1-F4
3. Multiscale exposure fusion (MEF)	Weights for contrast, saturation, saliency, exposure (Eqs. 7-11); Gaussian/Laplacian pyramids (Eqs. 12-14) in YUV	Final enhanced image

1. The pipeline at a glance

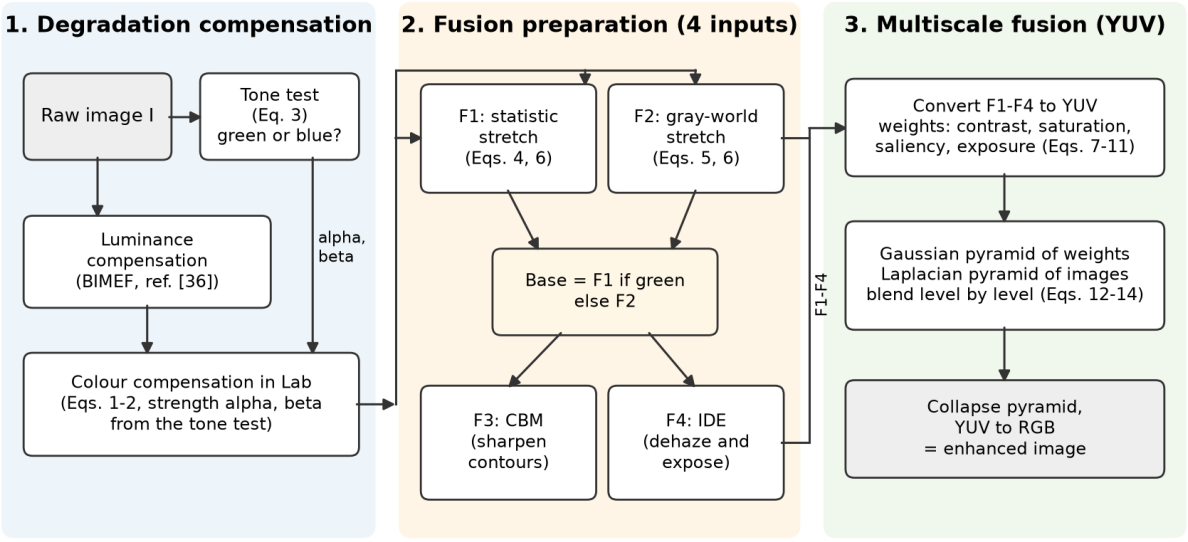


Figure 1. CDEF pipeline. The tone test controls both the colour compensation strength and which tone map is used as the base for F3 and F4.

Algorithm 1 of the paper, written as steps:

#	Step	Paper reference
1	Classify the raw image I as greenish if mean(G) > mean(B), otherwise bluish; set alpha, beta	Eq. 3
2	Luminance compensation: brighten under-exposed regions with BIMEF	Alg. 1 line 2, ref. [36]
3	Convert to CIELab; subtract a smoothed, masked share of a* and b* (strength alpha, beta); back to RGB	Eqs. 1-2
4	F1 = per-channel stretch to mean +/- 2 standard deviations	Eqs. 4, 6
5	F2 = per-channel stretch between gray-world-scaled quantiles	Eqs. 5, 6
6	Base = F1 if greenish, else F2	Alg. 1 lines 16-22
7	F3 = CBM(base): add morphological contour detail	ref. [6]
8	F4 = IDE(base): remove haze and correct exposure	ref. [40]
9	Fuse F1-F4 in YUV with weight maps and pyramids; convert back to RGB	Eqs. 7-14

2. Phase 1: Degradation compensation

2.1 Colour tone discrimination (Eq. 3)

What it does. Compares the average green and blue values of the raw image. If green is larger the image is *greenish*, otherwise *bluish*. The decision sets two parameters for colour compensation and decides which tone map later gets sharpened and dehazed.

greenish (mean G > mean B): alpha = 0.5, beta = 0.15 bluish: alpha = 1.0, beta = 0.3

Why this step

Underwater colour casts fall into these two groups, and they need different amounts of correction. Green is relatively well preserved underwater, so a greenish image needs a gentler correction; a bluish image has lost more and needs twice the strength.

Applying one fixed correction to every image is what makes many methods unstable across scenes: too weak on blue images, too strong (grey, washed out) on green ones. The test costs only two means.

alpha (for a*, the red-green axis) is larger than beta (for b*, the yellow-blue axis) because red is the colour that is lost most. The paper's Figs. 3-4 show that equal or larger beta turns images grey.

2.2 Luminance compensation with BIMEF

What it does. Brightens dark parts of the image while leaving well-lit parts alone. BIMEF (Ying et al., the paper's reference [36]) is itself a small exposure fusion:

- **Illumination map T.** Start from the brightest channel at each pixel, max(R, G, B), and smooth it while keeping strong edges (a weighted least-squares problem, solved on a half-size image). T tells how well lit each pixel is.
- **Synthetic exposure.** Make a brighter copy with a camera response model, $g(P, k) = e^{b(1 - k^a)} P^{k^a}$ (a = -0.3293, b = 1.1258). The exposure ratio k is chosen automatically to maximise the information (entropy) of the pixels that were under-exposed.
- **Blend.** $R = W P + (1 - W) g(P, k)$ with $W = T^{0.5}$: bright pixels keep the original, dark pixels take the brightened copy.

Why this step

Scattering and absorption lower all pixel values, not just colour. Colour compensation and contrast stretching both work better on an image whose dark regions have usable values; otherwise they amplify noise in near-black areas and give red artefact noise (the paper's Section III-B).

BIMEF is fast, has no training, and only brightens where needed, so it does not over-expose areas near artificial lights.

Evidence: removing this step ("w/o LC") lowered FDUM and CCF by about 9-11% in the paper and about 9% in our replication.

2.3 Colour compensation in CIELab (Eqs. 1-2)

What it does. Converts the brightened image to CIELab, where L* is lightness, a* runs from green (negative) to red (positive) and b* from blue (negative) to yellow (positive). A colour cast shows up as a* and b* being shifted away from zero. The method subtracts part of that shift:

$$I_a^c = I_a - \text{alpha} \cdot M \cdot G(I_a) \qquad I_b^c = I_b - \text{beta} \cdot M \cdot G(I_b)$$

- G is a Gaussian blur, so the correction follows the local cast smoothly instead of changing pixel by pixel (which would create artefacts).
- M is a mask that is 0 where the image is very bright (gray value above 0.85) and 1 elsewhere, so light sources and highlights keep their own colour (Eq. 2).
- This is the "3C" prior: in a natural image, the opponent colour channels average to about zero. Pulling them back toward zero removes the cast.

Why this step

Lab separates lightness from colour, so the colour can be corrected without changing brightness, and the a*/b* axes match the way underwater casts occur (one opponent pair is attenuated much more than the other).

Correcting only part of the shift (alpha, beta below 1 for greenish images) avoids the opposite mistake of a fully grey, colourless image.

Doing this before contrast stretching matters: a stretch applied to a blue image just makes a contrasty blue image.

3. Phase 2: Fusion preparation (four inputs)

The repaired image C still has low contrast and haze. Instead of one enhancement, CDEF makes four that each improve a different property, so the fusion can pick the best parts of each.

3.1 Two contrast stretches, F1 and F2 (Eqs. 4-6)

Both stretch each colour channel linearly so that a chosen low value maps to 0 and a chosen high value maps to 1 (Eq. 6), clipping what falls outside. They differ in how the limits are chosen:

$$\text{Eq. 6: } I_{tm} = (I - I_{min}) / (I_{max} - I_{min})$$

Input	Limits	Character
F1, statistic (Eq. 4)	mean - 2 std to mean + 2 std of the channel (a = 2)	Moderate, adapts to each channel's spread. Good for greenish images, where information is mostly kept.
F2, gray-world (Eq. 5)	Quantiles at b · (gray mean / channel mean) and 1 minus that (b = 0.01)	Stronger: a channel that is darker than the overall gray (usually red) gets its limits moved further in, so it is stretched more. Good for bluish images, which have lost more.

Why this step

Stretching restores the contrast that scattering removed and spreads the colour channels over the full range, which also corrects any remaining cast.

The paper averaged 4880 images into a mean greenish and a mean bluish image, tried both stretches on each, and found that the statistic stretch suits greenish images and the gray-world stretch suits bluish ones (its Fig. 5). Both go into the fusion; the tone decides which one is the base for F3 and F4.

Evidence: removing the tone mapping ("w/o TM") is the most damaging change in the ablation: CCF -17% in the paper, -25% in ours, and the colour cast stays (see Figure 5).

3.2 F3: Contour Bougie Morphology (CBM)

What it does. Sharpens object outlines using mathematical morphology (the authors' earlier method, reference [6]). Morphological opening removes small bright details; closing removes small dark ones. The difference between the image and its opened/closed version (a *top-hat*) is exactly the small bright and dark detail. CBM:

- Uses ring-shaped (contour) structuring elements of 15 x 15 and 5 x 5 pixels instead of filled squares, so it follows object contours.
- Repeats the open-close and close-open filters 10 times, collecting the white top-hat (bright detail) and black top-hat (dark detail) at each step.
- Adds the bright detail and subtracts the dark detail, scaled by omega = 0.1: F3 = base + 0.1 · (sum of white detail - sum of black detail).

Why this step

Forward scattering blurs edges and smears light onto object surfaces. Boosting bright and dark detail restores edges and textures (coral, sand, outlines) that a global stretch does not.

Morphology works on shapes rather than frequencies, so it sharpens contours with less ringing and noise amplification than an unsharp mask.

Evidence: "w/o CBM" is the mildest ablation (FDUM -2% in the paper, -6% in ours): CBM adds fine detail, which mostly shows in the sharpness-sensitive metrics.

3.3 F4: Image Dehazing and Exposure (IDE)

What it does. Removes the haze veil with a physical model (Ju et al., reference [40]). The standard haze model says each observed pixel is a mix of the true scene and the ambient water light A:

$$I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x))$$

where t(x) is the transmission (1 = clear, near 0 = all haze). IDE estimates A and t, then solves for the scene J. Its specific contributions are:

- **An enhanced scattering model** with an extra exposure term, so the result is not too dark (a known flaw of dark-channel dehazing).
- **Patch-wise transmission** from a closed-form quadratic, using local means over 75 x 75 patches, smoothed afterwards (a guided filter in the paper; it reduces to a box mean in the authors' code, which we follow).
- **Automatic strength.** The single free parameter t_{min} is found by search: the smallest value for which only a small share of pixels (epsilon = 0.02 above the minimum) would fall outside [0, 1]. Strong haze gives a low t_{min} and stronger dehazing.

Why this step

Back-scatter is a physical, depth-dependent veil; a contrast stretch cannot remove it because it treats near and far pixels the same. A model-based dehaze can.

IDE needs no training and sets its own strength per image, which fits CDEF's aim of stable results across scenes.

Its weakness (darkening, halos around bright objects) is limited because it is only one of four inputs: the fusion uses it where it helps (distant hazy water, see the F4 weight in Figure 3) and ignores it elsewhere.

3.4 Why apply CBM and IDE to the tone-matched base

Sharpening and dehazing amplify whatever is in the image, including colour errors. Running them on the stretch that suits the image's tone (F1 for greenish, F2 for bluish) gives them the best starting point. The paper also argues that general detail enhancement is enough here: an underwater-specific enhancer would add run time without a matching gain.

4. Worked example

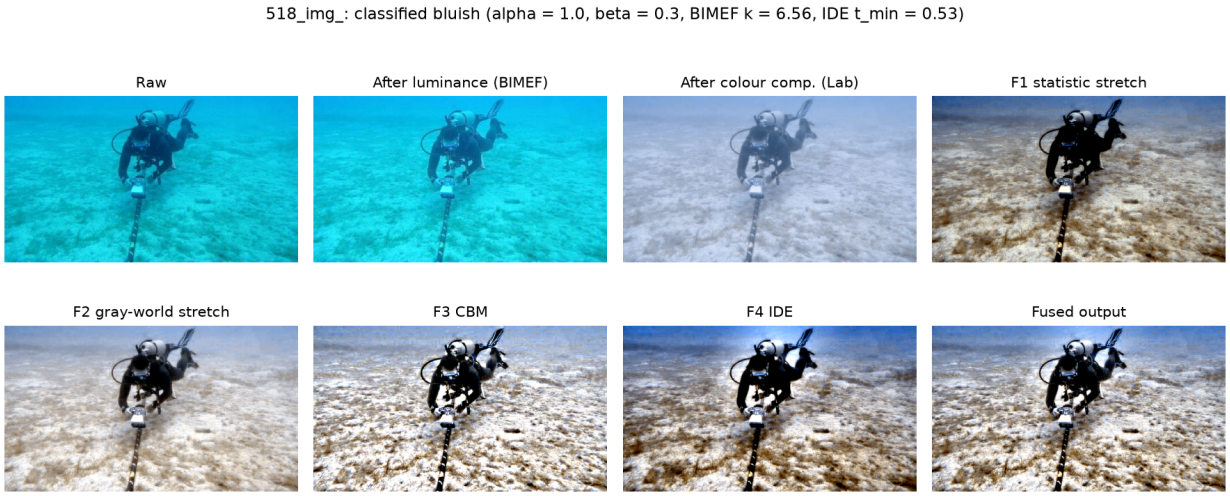


Figure 2. Every stage on a bluish UIEB image. Colour compensation removes the cyan cast; F1 is darker and more contrasty, F2 brighter; CBM sharpens texture; IDE removes the haze in the background but darkens; the fusion combines their strengths.

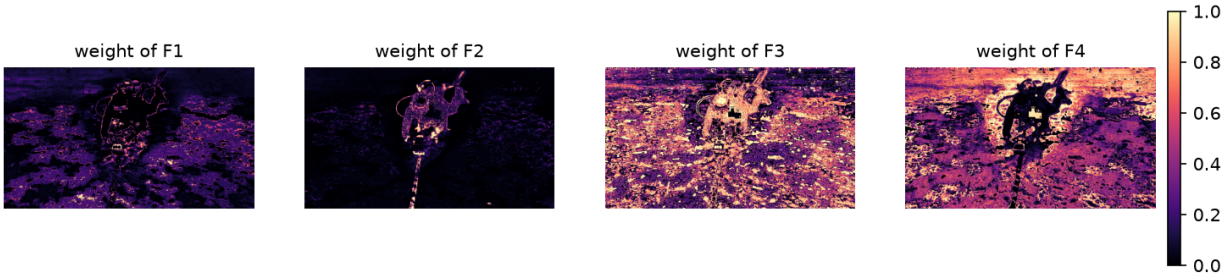


Figure 3. Normalised fusion weights for the same image (bright = trusted). F3 (CBM) dominates textured sand; F4 (IDE) dominates the hazy background water; F1 and F2 win on strong edges such as the diver's outline.

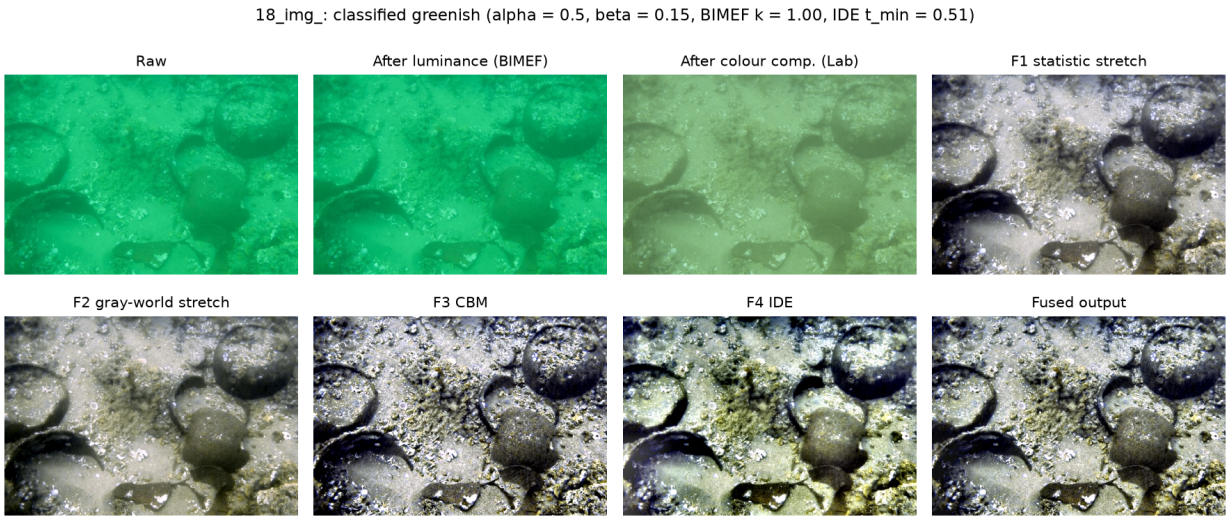


Figure 4. The same stages on a greenish image. Here F1 is the base for CBM and IDE.

5. Phase 3: Multiscale exposure fusion

5.1 Why fuse in YUV

YUV separates brightness (Y) from colour (U, V). Fusing in YUV lets the method measure contrast and exposure on the brightness channel alone and saturation on the colour channels alone, and blend exposure transitions smoothly in Y. In RGB every channel mixes brightness and colour, so a weight chosen for brightness also shifts colour.

5.2 The four weight maps (Eqs. 7-10)

Each input gets a weight at every pixel saying how good that input is there:

Weight	Formula	Rewards
Contrast W_c (Eq. 7)	|Laplacian of Y|	Edges and texture: sharp regions
Saturation W_s (Eq. 8)	$\|U\| + \|V\| + 1$	Vivid colour, against grey, washed-out regions
Saliency W_{sal} (Eq. 9)	$(Y - \text{mean } Y)^2 + (U - \text{mean } U)^2 + (V - \text{mean } V)^2$	Regions that stand out from the image average: the objects of interest
Exposure W_e (Eq. 10)	$\exp(-(Y - 0.5)^2 / (2 \cdot 0.2^2))$	Well-exposed pixels near mid-gray; penalises over- and under-exposure. This is CDEF's addition to the usual three underwater weights.

The four are multiplied and normalised so that, at each pixel, the weights of the four inputs add up to 1 (Eq. 11):

$$\bar{W}_k = (W_c \cdot W_s \cdot W_{sal} \cdot W_e)_k \,/\, \Sigma_j (W_c \cdot W_s \cdot W_{sal} \cdot W_e)_j$$

Why this step

Each input is good at something different (Figure 3). Per-pixel weights let the result take the dehazed version in the hazy background, the sharpened version on texture and the stretched versions on clean edges, instead of a single compromise everywhere.

Multiplying the measures means an input must be good on all of them to win a pixel: sharp but over-exposed pixels lose.

5.3 Pyramid blending (Eqs. 12-14)

Blending the inputs directly with these weights would create halos and seams, because weights change abruptly while the images do not. The method blends across scales instead:

- Each weight map is decomposed into a **Gaussian pyramid** (repeatedly blurred with a 5 x 5 kernel and halved).
- Each input is decomposed into a **Laplacian pyramid**: the detail lost between consecutive Gaussian levels, plus the coarsest level.
- At every level n, the result is $R_n = \Sigma_k G_n\{\bar{W}_k\} \cdot L_n\{F_k\}$ (Eq. 14). The pyramid is then collapsed back into one image.
- The number of levels adapts to image size: $n = \text{floor}(\log_2 \min(h, w))$ (Eq. 12), so large and small images are blended over their full range of scales.

Why this step

Fine detail is blended with sharp, local weights, while coarse brightness is blended with very smooth weights, so transitions between inputs are invisible. This is the standard exposure-fusion technique of Mertens et al. and is what the earlier underwater fusion methods use.

Evidence: fusing in RGB instead of YUV lowered CCF by 26% in the paper and 13% in our replication.

6. Justification summary

Each component targets one kind of degradation, and the ablation study (paper's Table IV) measures what is lost without it. CCF measures colourfulness, contrast and fog density; FDUM is an underwater quality measure that weighs colour, contrast and sharpness.

Component	Degradation it addresses	CCF without it (paper / ours)	FDUM without it (paper / ours)
Tone test + colour compensation	Blue/green colour cast, lost red	not ablated in the paper	-
Luminance compensation (BIMEF)	Dark, under-exposed regions	-10.8% / -8.7%	-9.7% / -8.7%
Tone mapping F1, F2	Low contrast, remaining cast	-17.3% / -25.3%	-15.9% / -19.9%
CBM (F3)	Blurred contours (forward scatter)	-1.8% / -2.0%	-2.3% / -6.1%
IDE (F4)	Haze veil (back-scatter)	-23.9% / -3.2%	-15.2% / -3.9%
Fusion in YUV (vs RGB)	Seams, halos, colour shifts from blending	-26.4% / -13.1%	-13.6% / -6.8%

Every ablated version scores below the full method on URanker, a learned quality ranker, in both the paper and our replication. Luminance compensation, tone mapping and CBM reproduce closely. The IDE and RGB rows drop less in ours. We ran the authors' own released code with the same removals and it also drops only 4.6% (IDE) and 3.5% (RGB) on CCF, so those two paper numbers were probably produced with a different version of the code.

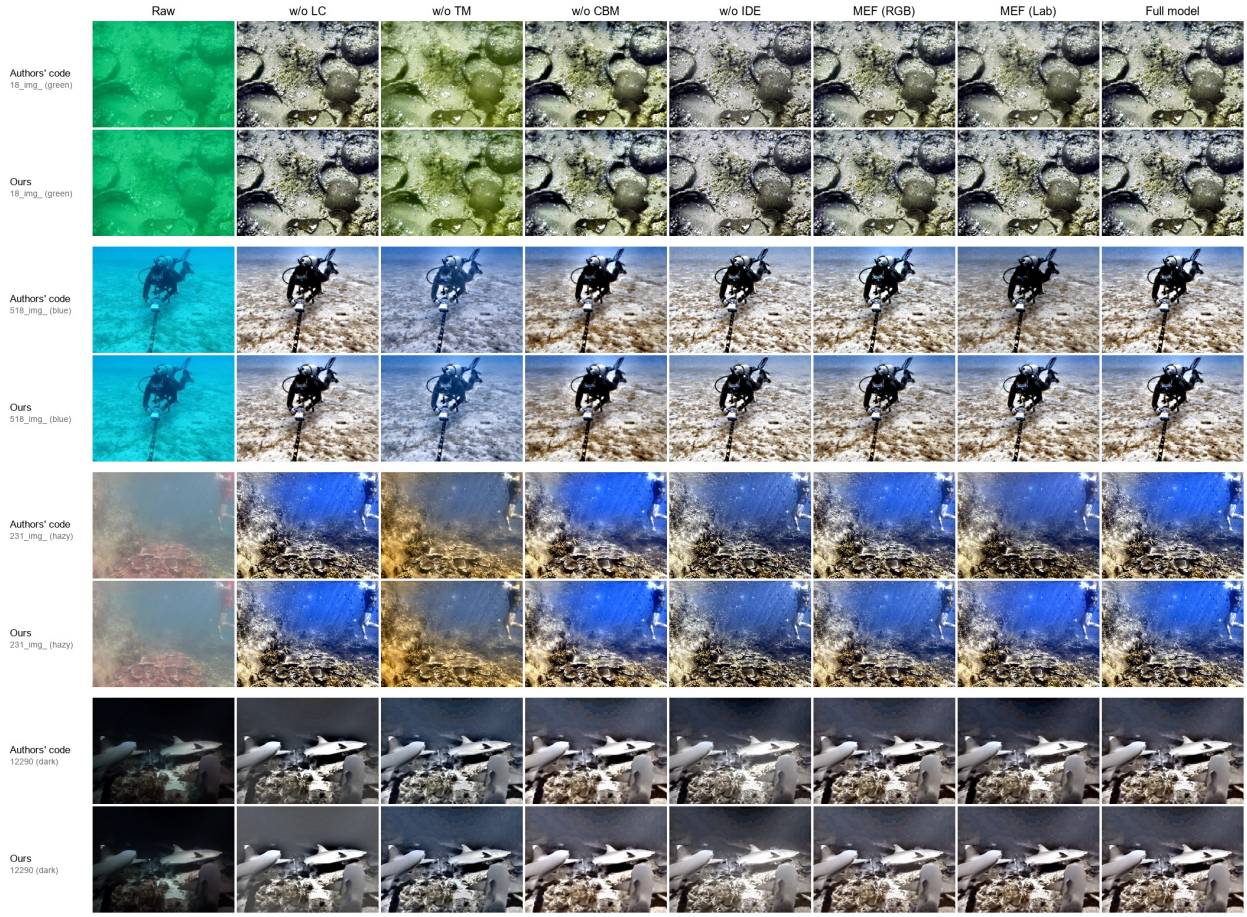


Figure 5. Ablation outputs from the authors' code and ours. Without tone mapping the cast remains; without luminance compensation the image is flatter; the other removals change fine detail rather than the overall look.

Design principles behind the method

- **Compensate before enhancing.** Fixing brightness and colour first makes every later step behave the same on very different images, which is the paper's main claim (stable performance).
- **Adapt to the scene with a cheap decision.** One comparison of two channel means switches the correction strength and the base for detail enhancement.
- **Many partial fixes, one per-pixel choice.** No single enhancement is right everywhere; weighted fusion keeps each where it is best and hides its side effects elsewhere.
- **No training data.** Underwater images have no true clean reference, so a learning method's labels are themselves imperfect enhancements. CDEF needs no labels and runs on a CPU.

7. Our implementation

Everything was written from scratch in Python (NumPy, OpenCV, SciPy), from the papers. The authors' released MATLAB code was read afterwards only to explain gaps, and each choice we took from it is a setting, with the paper's reading kept as an option (configs/cdef_paper.yaml, docs/deviations.md). Where the paper is ambiguous or inconsistent, we chose as follows:

Paper says	Problem	Our choice
Eq. 10: phi = 2, sigma = 0.5	Text says well-exposed is 0.5; phi = 2 would reward the brightest pixels	phi = 0.5, sigma = 0.2 (Mertens et al., matches the text)
Eq. 11: (W + eps) / (sum W + eps), eps = 0.1	Does not sum to 1 as the text requires; eps = 0.1 flattens the weights into a plain average	Weights normalised to sum to 1, eps = 1e-12 (authors' code)
Eq. 7 Laplacian [0 1 0; -1 4 -1; 0 -1 0]	Typo: not a Laplacian	Standard [0 1 0; 1 -4 1; 0 1 0]
Eq. 4: "variance"	Units do not match the mean	Standard deviation
Weight exponents all 1	The authors' code squares contrast and saliency	Squared, as in their code (affects sharpness of the blend)
Brightness of each input not mentioned	The authors' fusion (Wang et al.) scales weights by mean(Y)^2	Included, as in their code
BIMEF, CBM and IDE given only by reference	Settings not stated	Values from the cited papers and released code (e.g. IDE strength search, BIMEF half-size solve)

Where the paper's own rule and the authors' code differed but the scores changed by less than 1% on a fixed 80-image sample, we kept the paper's rule (for example, fusion in YUV rather than their YCbCr, and the full pyramid depth of Eq. 12).

8. Results

All scores use the metric authors' own code, on the full datasets. Raw-image scores match the paper's Table I to within 0.02% on all four datasets, which confirms the same images and metric code.

Dataset	UCIQE	FDUM	CEIQ	CCF	URanker
UIEB (890), ours	0.6525	0.9122	3.5760	48.22	2.4658
UIEB, paper	0.6468	0.8307	3.5544	45.11	2.3990
C60 (60), ours	0.6367	0.6397	3.5119	38.39	1.9241
C60, paper	0.6315	0.5832	3.4867	36.04	1.8214
EUVP (515), ours	0.6447	0.7762	3.5689	63.15	3.3582
EUVP, paper	0.6367	0.7062	3.6076	57.81	3.3213
UCCS (300), ours	0.6160	0.7033	3.6317	47.48	2.6746
UCCS, paper	0.6077	0.5925	3.6328	43.74	2.5748

Ours is within about 1.5% of the paper on UCIQE and CEIQ and above it on FDUM, CCF and URanker. The authors' released code also scores above their paper (UIEB CCF 46.56), so part of the gap comes from the paper itself. Higher scores do not always mean better-looking images.

Metrics used

Metric	What it measures
UCIQE	Underwater colour image quality: chroma spread, saturation and luminance contrast
FDUM	Frequency-domain underwater metric: colourfulness, contrast and sharpness
CEIQ	Quality of contrast-distorted images (entropy and histogram based)
CCF	Colourfulness, contrast and fog density combined: rewards haze removal
URanker	A transformer network trained to rank underwater images by quality

Limitations

- The detail inputs can over-shoot: halos can appear around bright objects (IDE) and noise in flat areas (CBM), which the fusion only partly hides.
- The tone test is binary; images with mixed or weak casts get one of two fixed strengths.
- Several settings (alpha, beta, a, b, omega) are empirical, chosen on the paper's datasets.
- Run time in Python is about 1-3 s for a typical 640-1280 px image; BIMEF and IDE take most of it.